



Brain Tumor Segmentation

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Motivation for Your Project

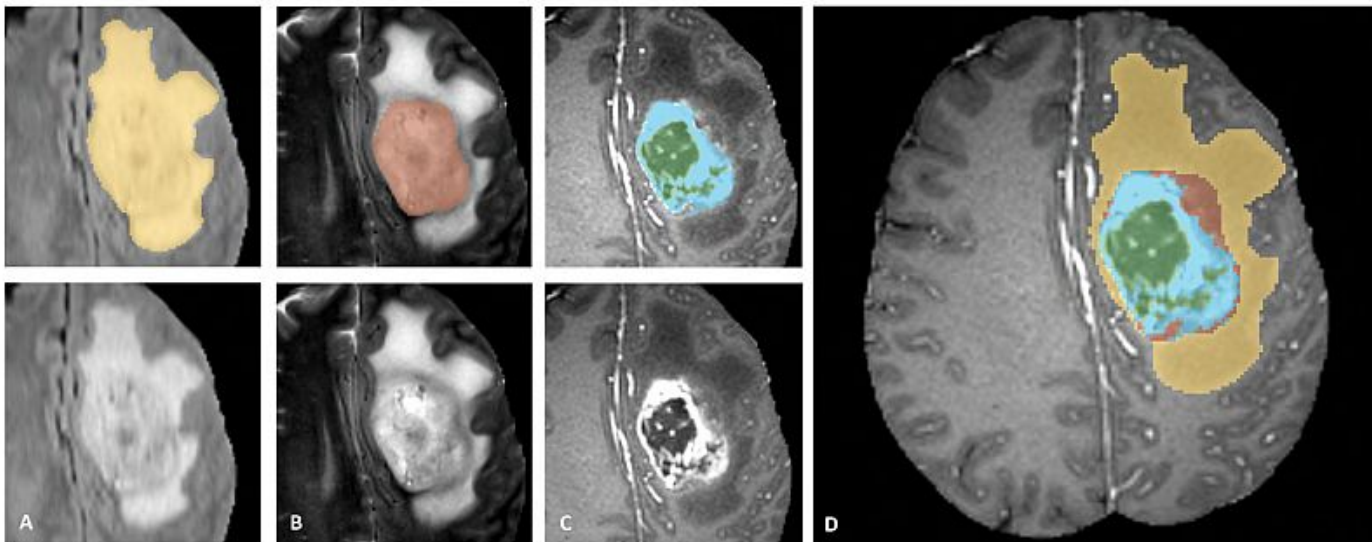
S - Situation

T - Task

A - Action

R - Result

- Brain tumors are a deadly disease
- Accurately detecting tumors is important for treatment and prognosis
- Solution: segment glioma tumors with deep learning





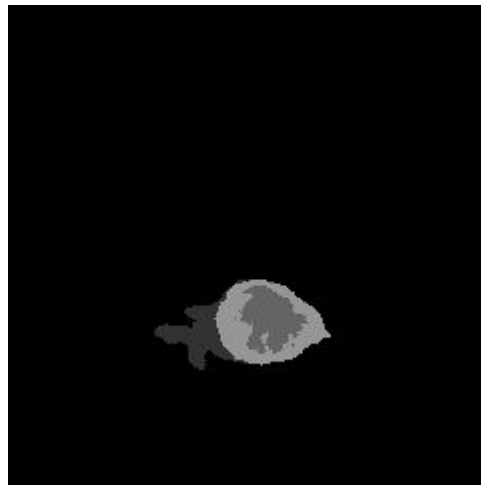
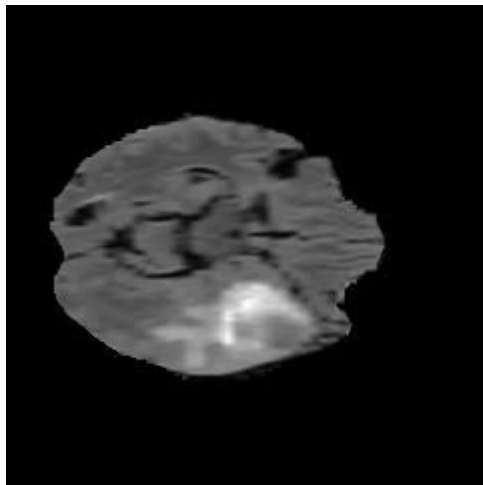
Your Data

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- 4715 MRI images
- 152 patients with glioma



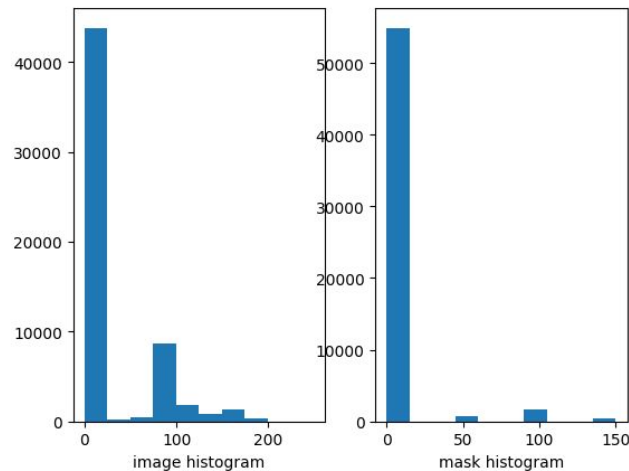
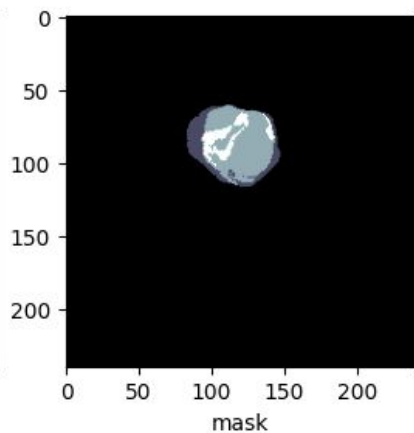
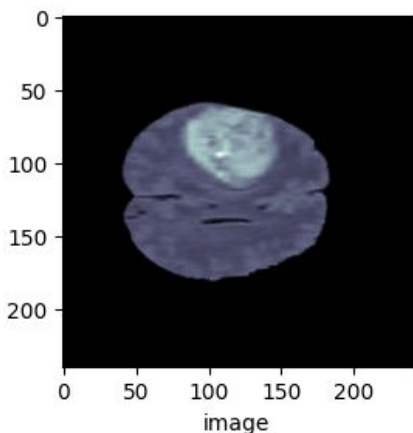
Exploratory Data Analysis

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MRI

- Minimum = 0
- Maximum = 249
- Mean = 24
- Standard deviation = 45

Mask

- Minimum = 0
- Maximum = 150
- Mean = 4
- Standard deviation = 21
- Unique = 0, 50, 100, 150



Preprocessing the Data

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1. Open
2. Decode
3. Binarize classes
4. Resize
5. One-hot encode

```
def preprocess_data(image_path, mask_path, input_shape, num_classes):  
    image = tf.io.read_file(image_path)  
    mask = tf.io.read_file(mask_path)  
    image = tf.image.decode_jpeg(image, channels=1)  
    mask = tf.image.decode_png(mask, channels=1)  
    image = tf.image.convert_image_dtype(image, tf.float32)  
    mask = tf.math.greater(mask, 0)  
    mask = tf.cast(mask, tf.int32)  
    image = tf.image.resize(image, (input_shape[0], input_shape[1]))  
    mask = tf.image.resize(mask, (input_shape[0], input_shape[1]), method=tf.image.ResizeMethod.NEAREST_NEIGHBOR)  
    mask = tf.one_hot(mask, depth=num_classes)  
    mask = tf.squeeze(mask, axis=-2)  
    return image, mask
```



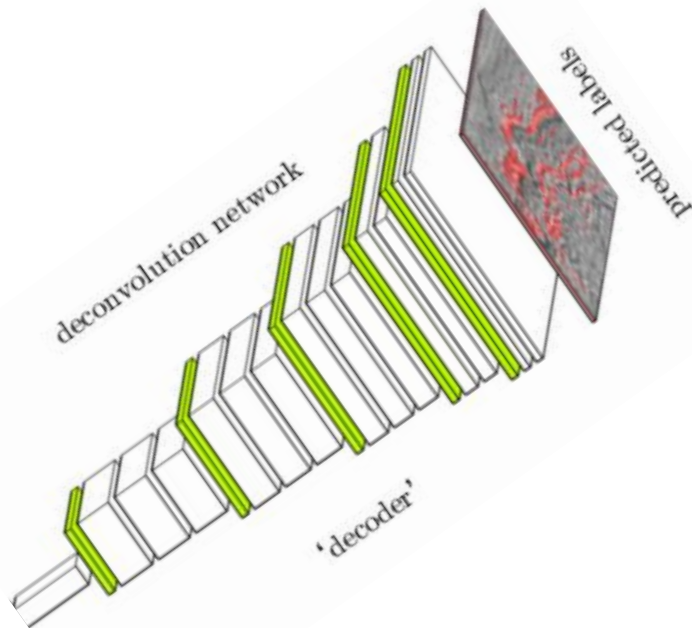
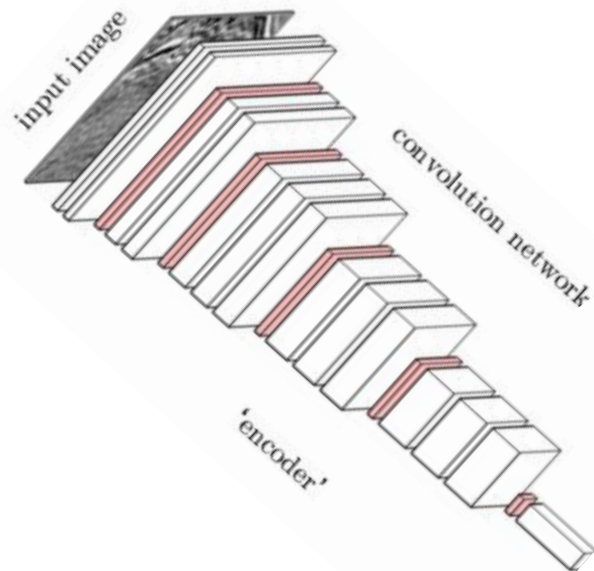
Baseline Model and Results

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U-Net Architecture



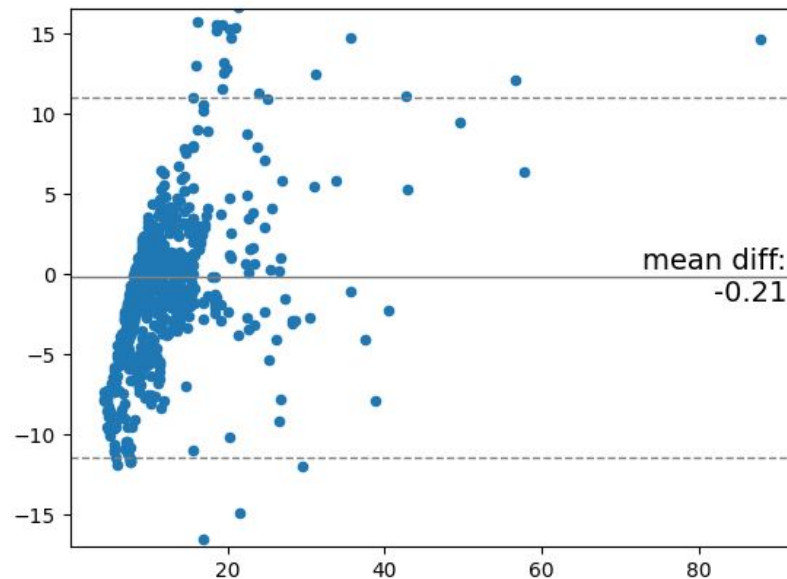
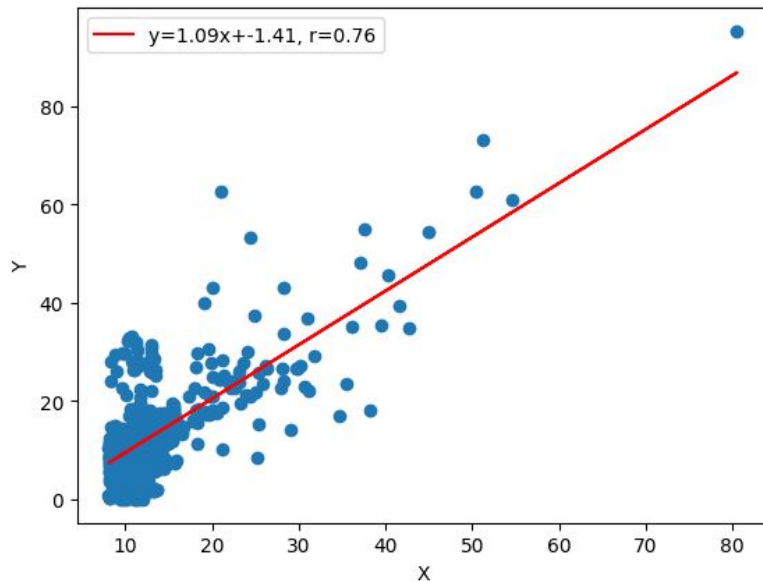
Baseline Model and Results

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Project Summary and Future Work

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Summary

- Developed an initial U-net for segmenting brain tumors from MRI images and computing tumor volumes
- Reasonable performance: $r = 0.76$
- Larger tumors tend to be predicted larger than they really are

Future Work

- Create a more advanced model to improve our accuracy
 - Continue model training (more epochs)
 - Train additional models with different architectures



Thanks for Listening!